

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: Power BI And Tableau Practical

PRACTICAL - 2

Aim:- Working with Data Sources and Data Preparation in Tableau

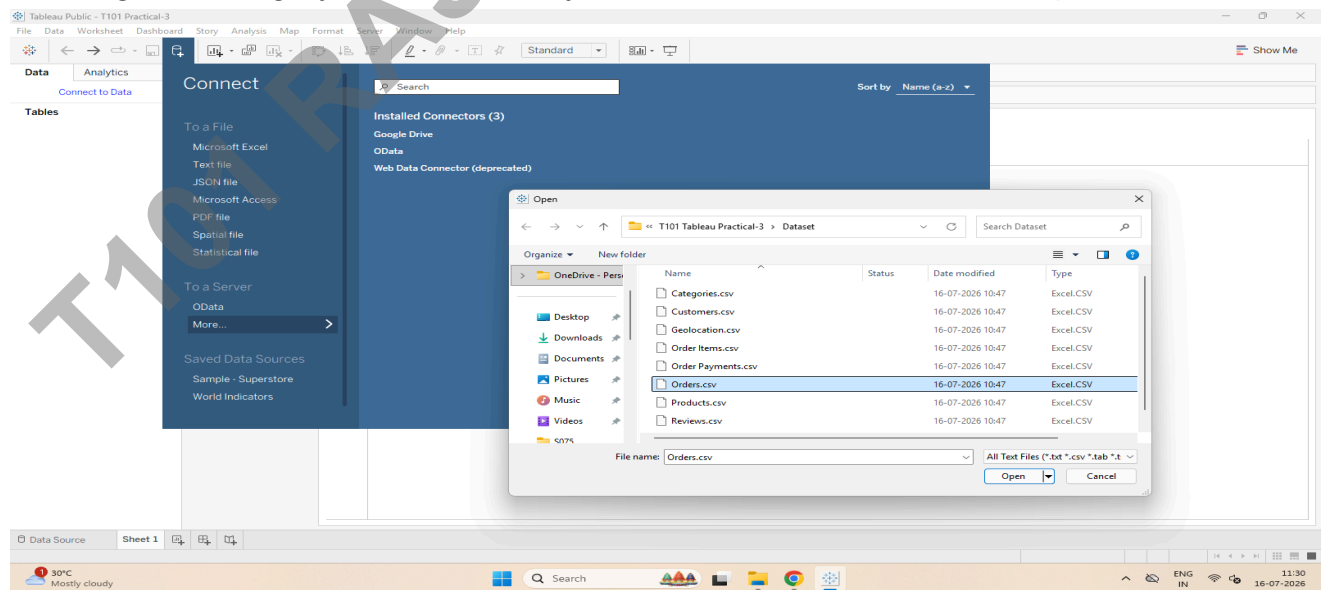
- A. Connect to different data sources (Excel, CSV, Database)
- B. Create and use data extracts vs live connections
- C. Manage metadata (rename, hide, change data types)
- D. Perform joins (inner, left, right) on multiple tables
- E. Apply data blending between two sources
- F. Implement different types of filters (dimension, measure, date)
- G. Optimize performance using extracts and filtering

Dataset: E-commerce Dataset

Steps:-

Step A: Connecting to Different Data Sources

1. Launch the Tableau Desktop application.
2. Locate the blue Connect control panel on the far-left section of the start page.
3. Under the To a File sub-header, click on Text File (Tableau classifies .csv spreadsheet extensions as flat text files).
4. Navigate through your file directory, select Orders.csv, and click Open.

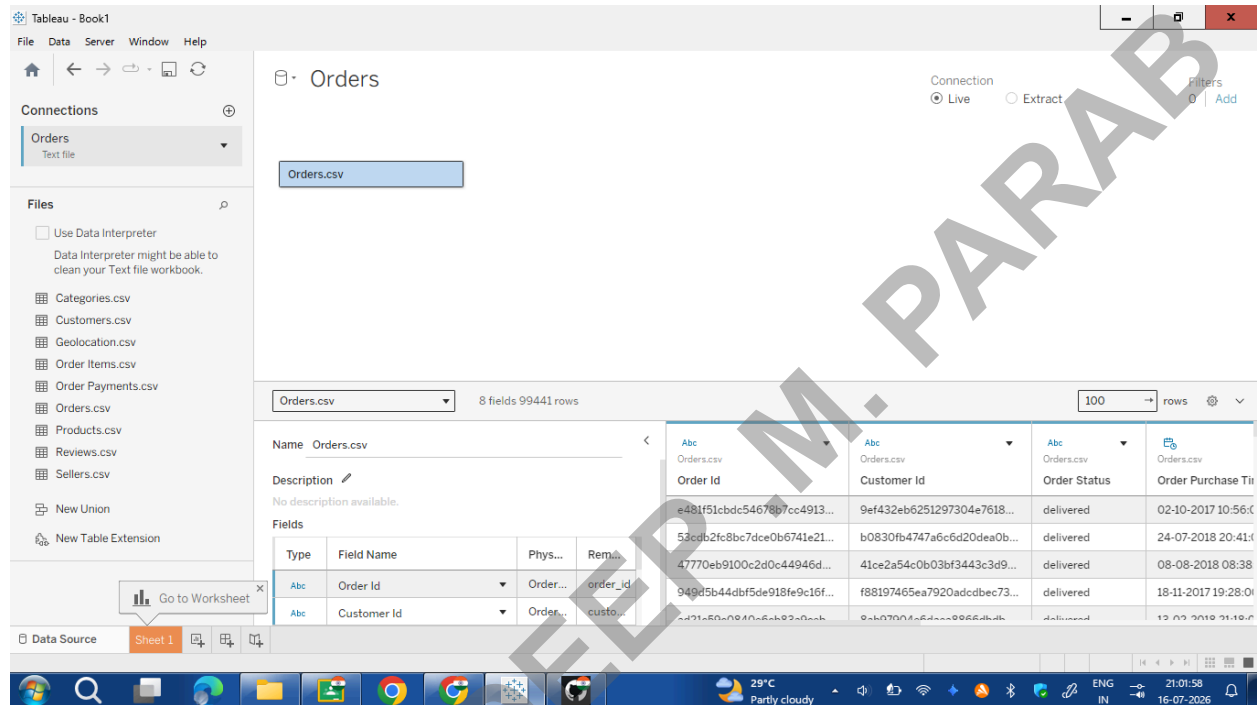


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Step B: Creating Data Extracts vs. Live Connections

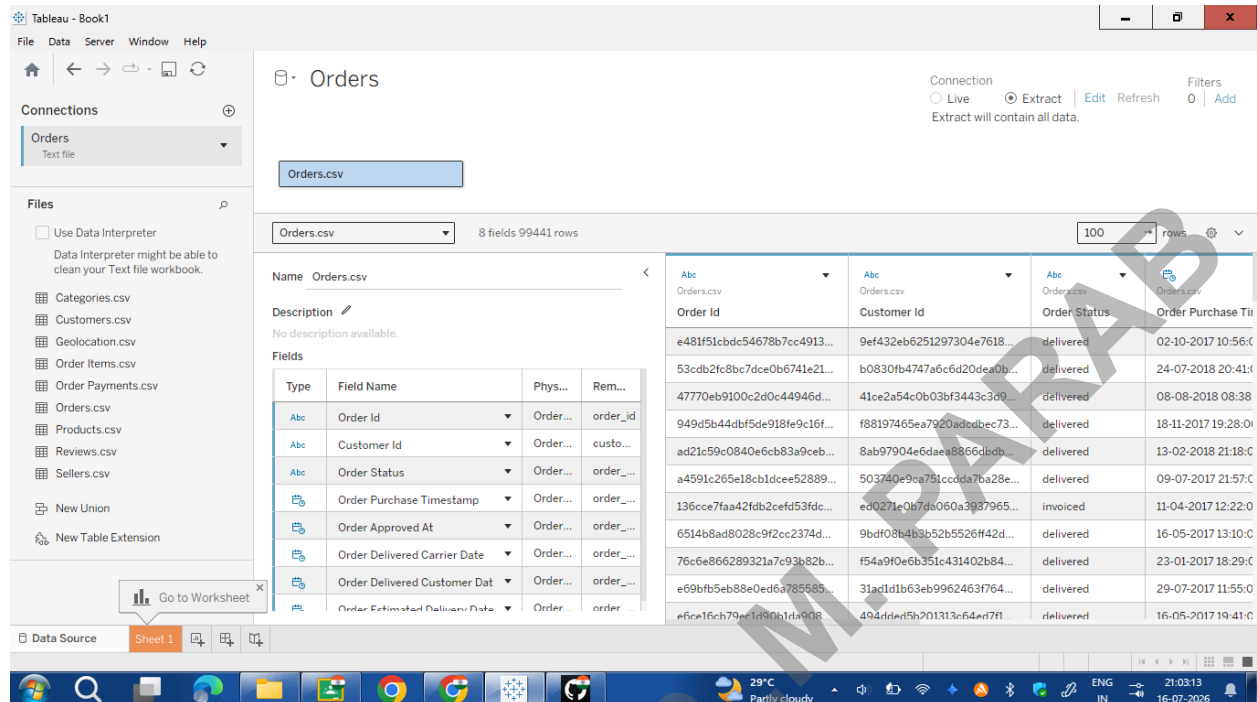
1. Once the dataset populates the main staging view, navigate to the top-right corner of the Data Source interface.
2. Locate the connection mode toggle section



3. Switch the active selection from Live over to Extract.

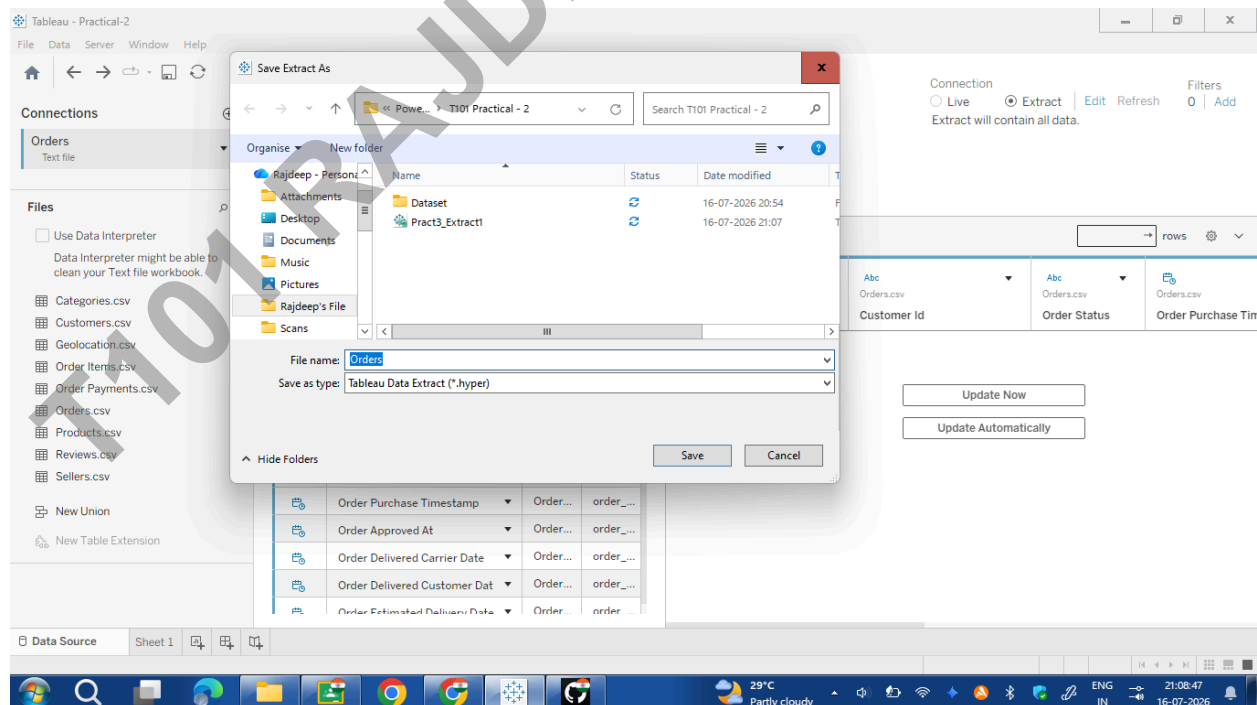
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4. Click on the Sheet 1 navigation tab located at the bottom-left corner of the workspace screen.

5. A system dialogue save window will immediately appear. Name the local snapshot file Ecommerce_Extract and click Save.



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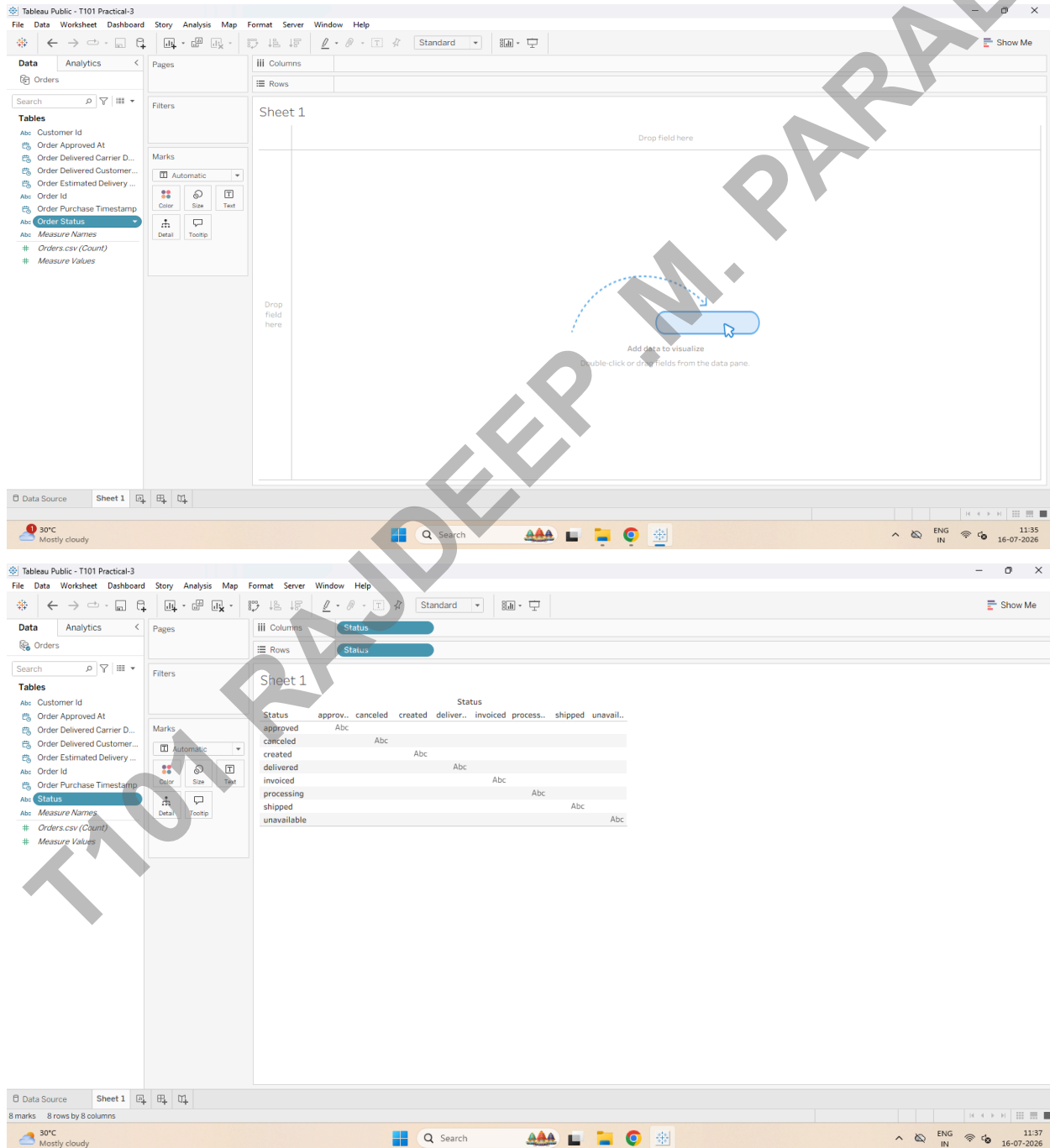
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Step C: Managing Operational Metadata

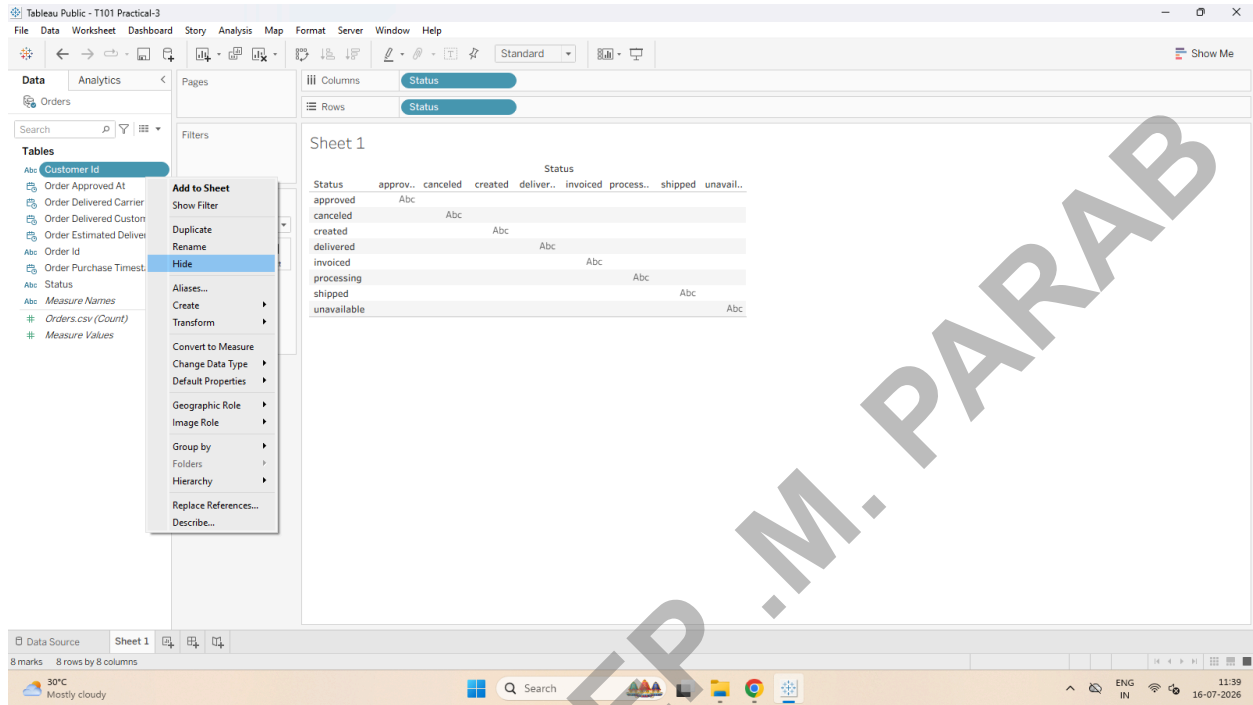
1. Navigate to the blank workspace canvas on Sheet 1 and view the left-hand side Data Pane.
2. Rename a Field: Locate the dimensional attribute field labeled order_status or Order Status. Right-click the field name, select Rename, type in Status, and hit Enter.



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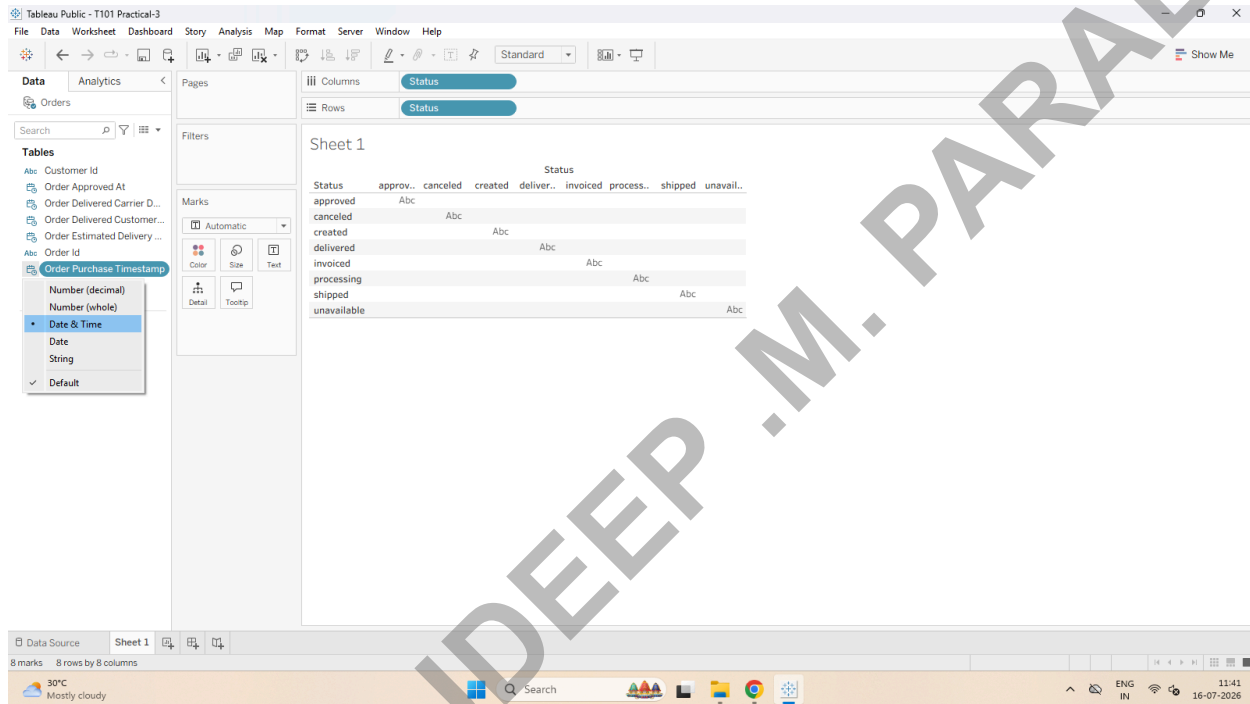
3. Hide a Field: Locate an internal tracking string variable that will not be used in visual reporting, such as customer. Right-click the field name and select Hide.



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4. Change a Data Type: Locate the operational timestamp column labeled order_purchase_timestamp. Click directly on the data type icon (which may incorrectly display as text Abc or a calendar date) positioned immediately to the left of the field name. Select Date & Time from the context dropdown menu to force proper structural recognition (Note: It may by default selected to Date and Time).



Step D: Performing Relational Joins on Multiple Tables

1. Click the Data Source tab positioned at the bottom-left corner of the window to return to the data schema designer.
2. In the top staging canvas area, locate the rectangular Orders.csv database block.
3. Double-click directly on the Orders.csv block. This action drills down through the logical relationship layer and enters the Physical Layer canvas space.
4. From the left-hand files listing pane, click and drag the Order Payments.csv file, dropping it directly next to the active Orders.csv block. A traditional Venn diagram icon link will snap open between them.

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Tableau Public - T101 Practical-3

File Data Window Help

Connections

Orders
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Orders

Orders.csv

8 fields 99441 rows

Name Orders.csv
Description
No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	Order Status	Order...	order_...
Abc	Order Purchase Timestamp	Order...	order_...

Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approved At
e481f51cdbc54678b7cc4913...	9ef432eb6251297304e7618...	delivered	02-10-2017 10:56:00	02-10-2017 11:07:00
53cdb2fc8bc7dce0b6741e21...	b0830fb4747a6c6d20dea0b...	delivered	24-07-2018 20:41:00	26-07-2018 03:24:00
47770eb9100c2d0c44946d...	41ce2a54c0b03bf3443c3d9...	delivered	08-08-2018 08:38:00	08-08-2018 08:55:00
949d5b44dbf5de918fe9c16f...	f88197465ea7920adcdbe73...	delivered	18-11-2017 19:28:00	18-11-2017 19:45:00
ad21c59c0840e6cb83a9ceb...	8ab97904e6dae8866dbdb...	delivered	13-02-2018 21:18:00	13-02-2018 22:20:00
a4591c265e18cbidcee52889...	503740e9ca751cedda7ba28e...	delivered	09-07-2017 21:57:00	09-07-2017 22:10:00
136cce7aa42fdb2cef053fde...	ed0271e067da060a3937965...	invoiced	11-04-2017 12:22:00	13-04-2017 13:25:00

Data Source Sheet 1

Tableau - Practical-2

File Data Server Window Help

Connections

Orders
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Orders

Orders.csv

13 fields 103886 rows

Name Orders.csv
Description
No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...

Orders.csv is made of 2 tables

Orders.csv

Order Payments.csv

Connection
☐ Live ☒ Extract | Edit Cancel
Extract will contain all data.

Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approved At
916be5eed...	5d51e1af408ce...	delivered	13-05-2018 19:36:00	13-05-2018 19:...
5308fdab5...	3aec6e3846fd...	delivered	02-06-2018 11:51:00	02-06-2018 12:...
11c177c8e...	b331b74b18dc...	delivered	08-01-2018 21:28:00	08-01-2018 21:...
a61574f3d...	b87bbd15537c...	delivered	03-03-2018 22:19:00	03-03-2018 22:...

Data Source Sheet 1

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- Click directly on the Venn diagram icon to open the Join configuration properties box.
- Under the data grid coupling properties, select order_id as the matching data key column drop-down for both the left and right tables (Note: in some case Tableau automatically selects the Order Id)
- Toggle Join Configurations:-
Click Inner: Observe the underlying data preview window at the bottom. The rows instantly filter to show only records that contain both an order entry and a corresponding transaction payment receipt.

The screenshot shows the Tableau interface with a join configuration window open. The join is set to Inner. The data preview shows columns: Order Id, Customer Id, Order Status, Order Purchase Timestamp, and Order Approver. The data preview shows the following data:

Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approver
916be5eed...	5d51e1af408ce...	delivered	13-05-2018 19:36:00	13-05-2018 19:36:00
5308fdab5...	3aec6e3846fd...	delivered	02-06-2018 11:51:00	02-06-2018 12:00:00
11c177c8e...	b331b74b18dc...	delivered	08-01-2018 21:28:00	08-01-2018 21:28:00
a61574f3d...	b87bbd15537c...	delivered	03-03-2018 22:19:00	03-03-2018 22:19:00
77567a06...	shaco07d4079...	cancelled	27-08-2018 10:54:00	27-08-2018 10:54:00

Click Left: Notice the view updates to safely preserve 100% of the root rows coming from Orders.csv, leaving payment parameter cells completely empty if an order has no matching payment record.

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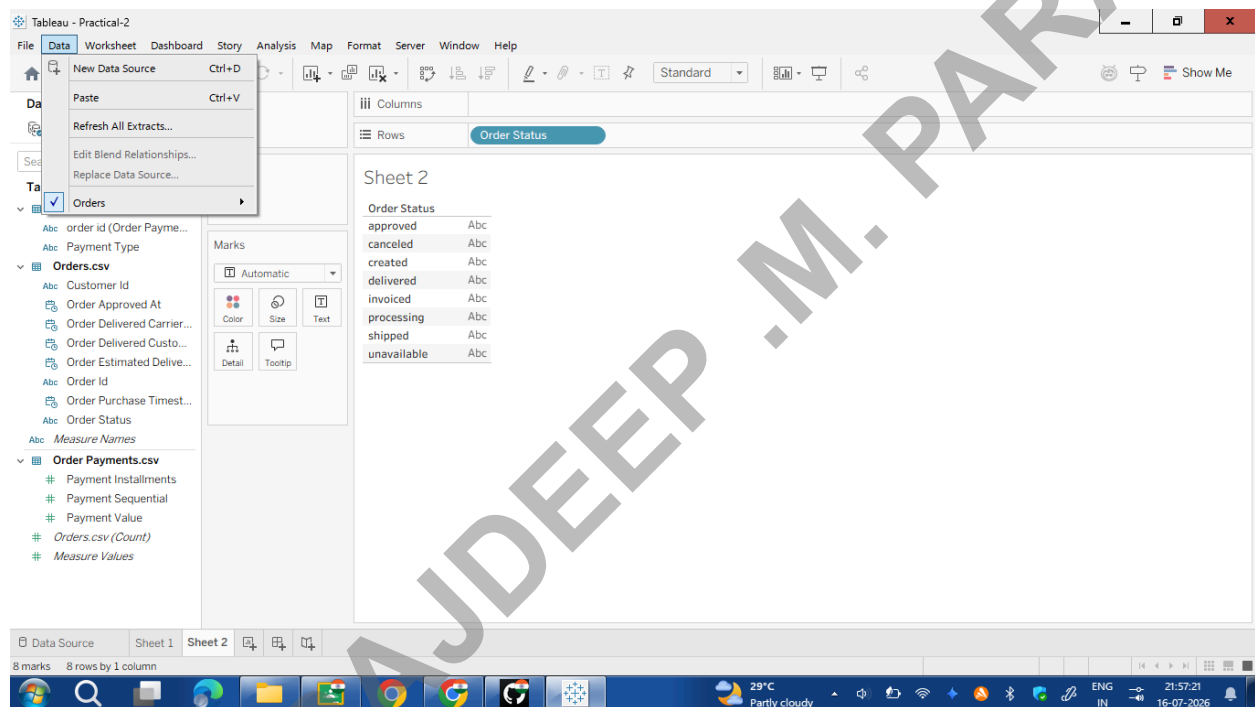


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Step E: Applying Data Blending Between Independent Sources

1. Create a brand new blank workspace sheet by clicking on the New Worksheet icon (the chart block containing a \$\$ symbol) positioned at the bottom navigation tab area. This opens Sheet 2.
2. Ensure your joined Orders dataset connection is highlighted at the top left of the Data Pane. Drag the newly modified Status dimension field and drop it onto the Rows shelf. A blue checkmark will appear on the source, designating it as the Primary data pipeline.

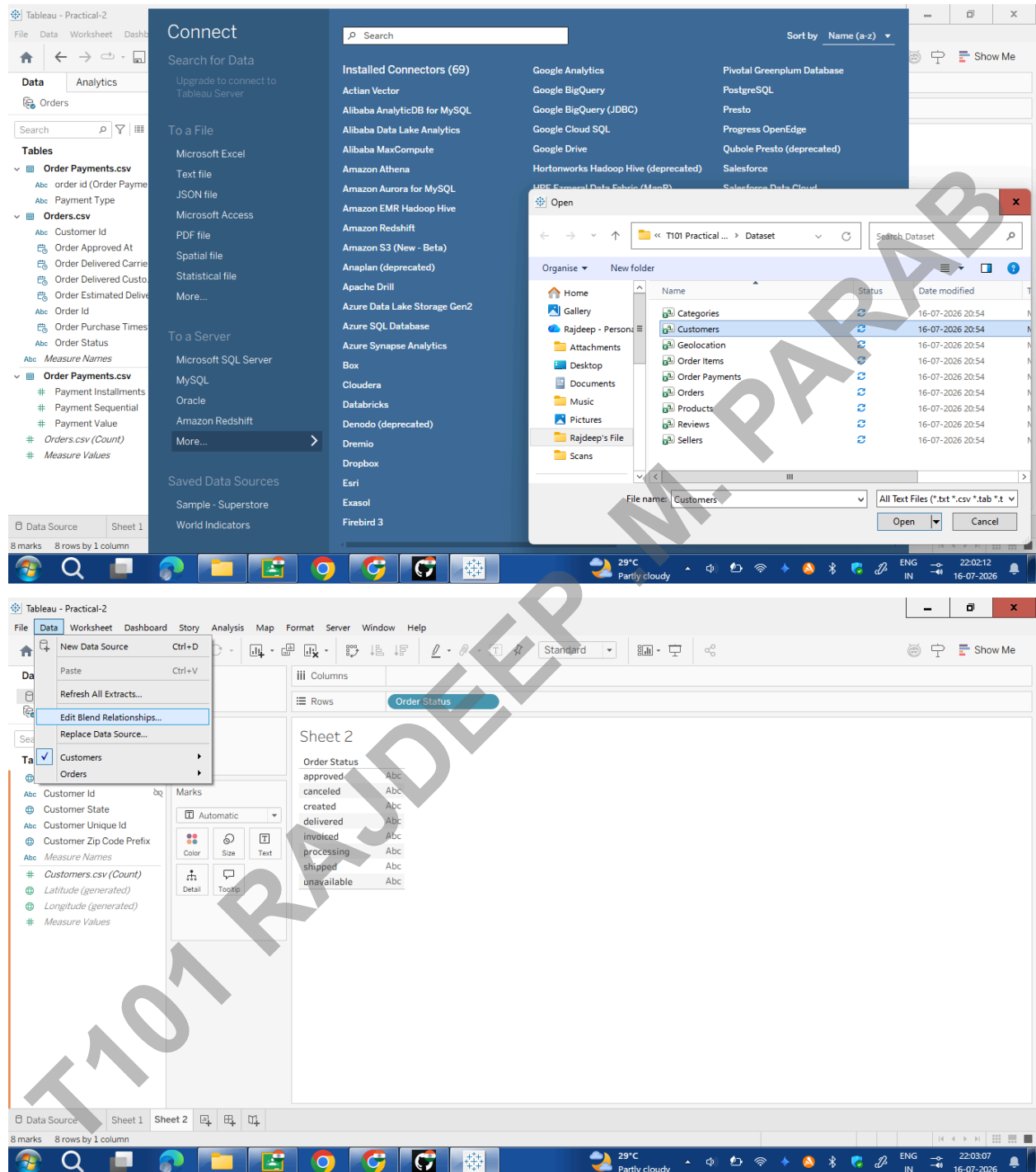


After Clicking new data source you are directed to blue menu window select text file to add the Customers.csv file.

After adding the CSV file get back to sheet 2 and recheck the Edit Blend Relationship Option and click on it .

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4. Set the Primary data source field box to your active Orders connection, and map the Secondary data source block to the independent Customers dataset file.

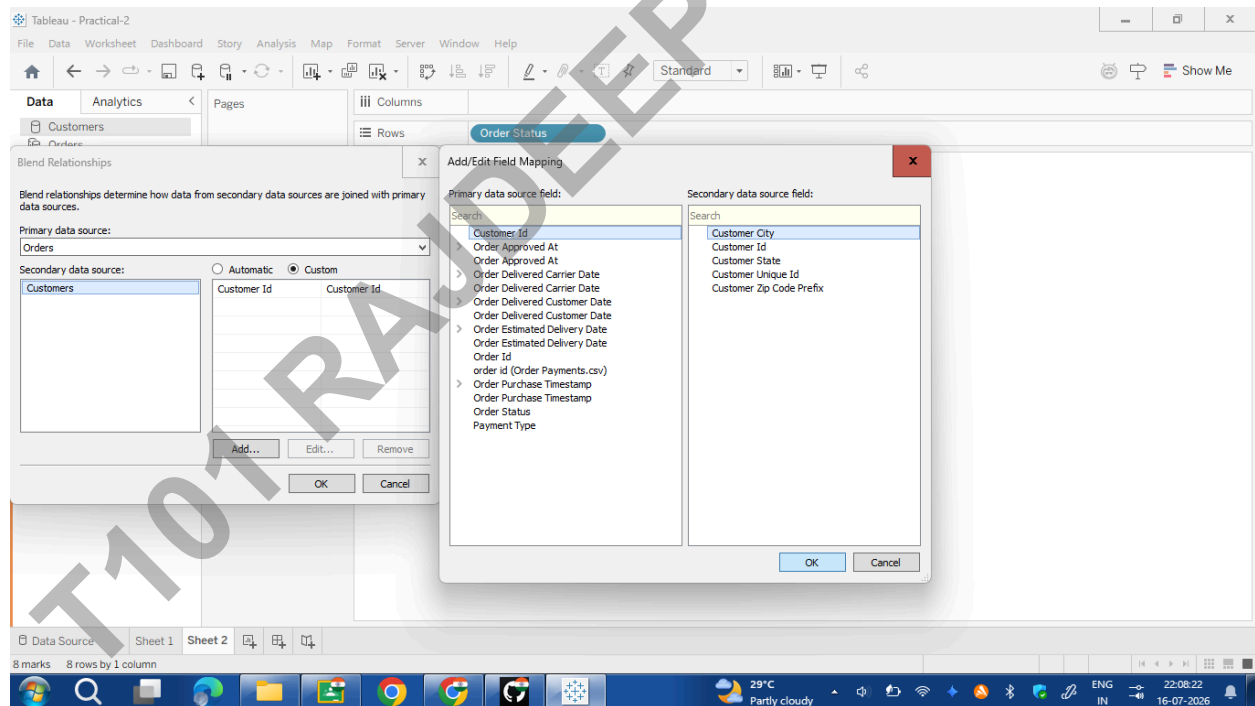
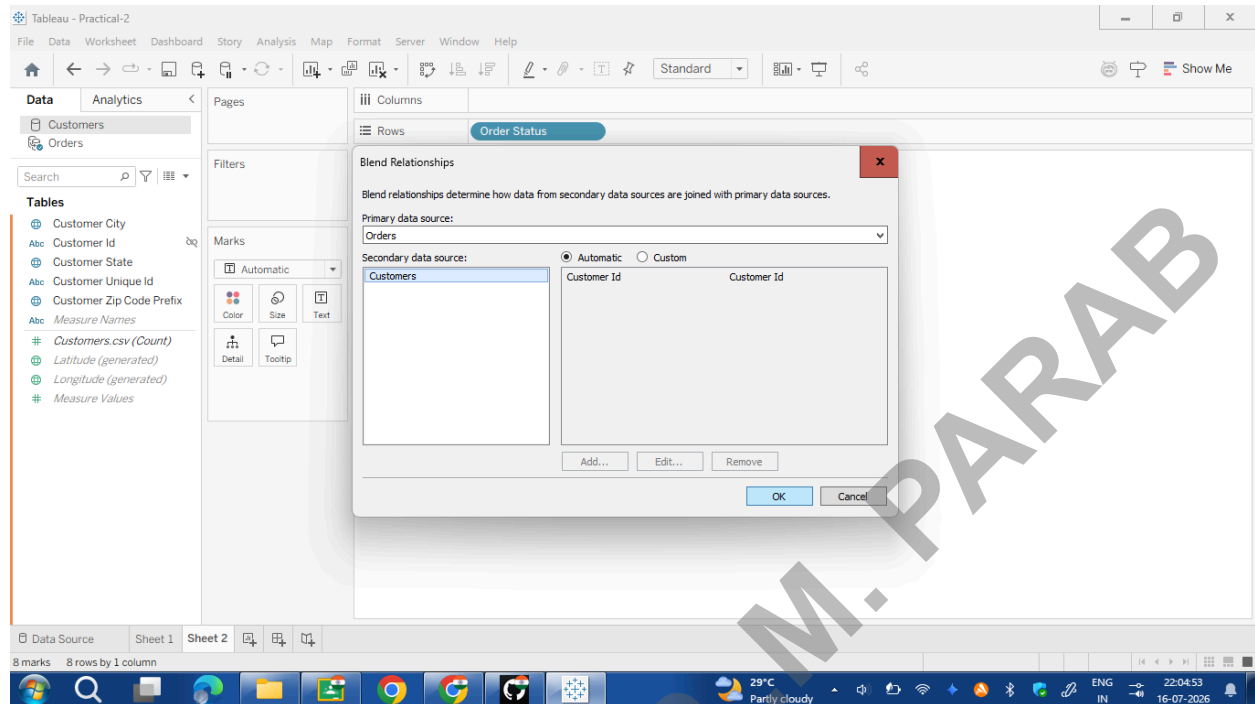
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If the Customer Id is not visible in primary data its not an issue since we hided it earlier at Step C

Now Navigate Back to data source and change the connection to “LIVE” from

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“EXTRACT”

The screenshot displays the Tableau Desktop interface for a practical exercise. The main window is titled 'Tableau - Practical-2'. The 'Connections' pane on the left shows 'Orders' as a 'Text file'. The 'Files' pane lists various CSV files, including 'Orders.csv'. The main view shows the 'Orders' data source, with a logical table named 'Orders.csv' and physical tables 'Orders.csv' and 'Order Payments...'. A data preview table is shown with the following columns: Order Id, Customer Id, Order Status, and Order Purchase Timestamp. The table contains 10387 rows of data.

Order Id	Customer Id	Order Status	Order Purchase Timestamp
916be5eed...	5d51e1af408cef36611d1ea5d...	delivered	13-05-2018 19:36:00
5308fdab5...	3aee6e3846fd122ac430ac1a...	delivered	02-06-2018 11:51:00
11c177c8e...	b331b74b18dc79bcd6532d5...	delivered	08-01-2018 21:28:00
a61574f3d...	b67bbd15937c8fdb8f0f378f...	delivered	03-03-2018 22:19:00

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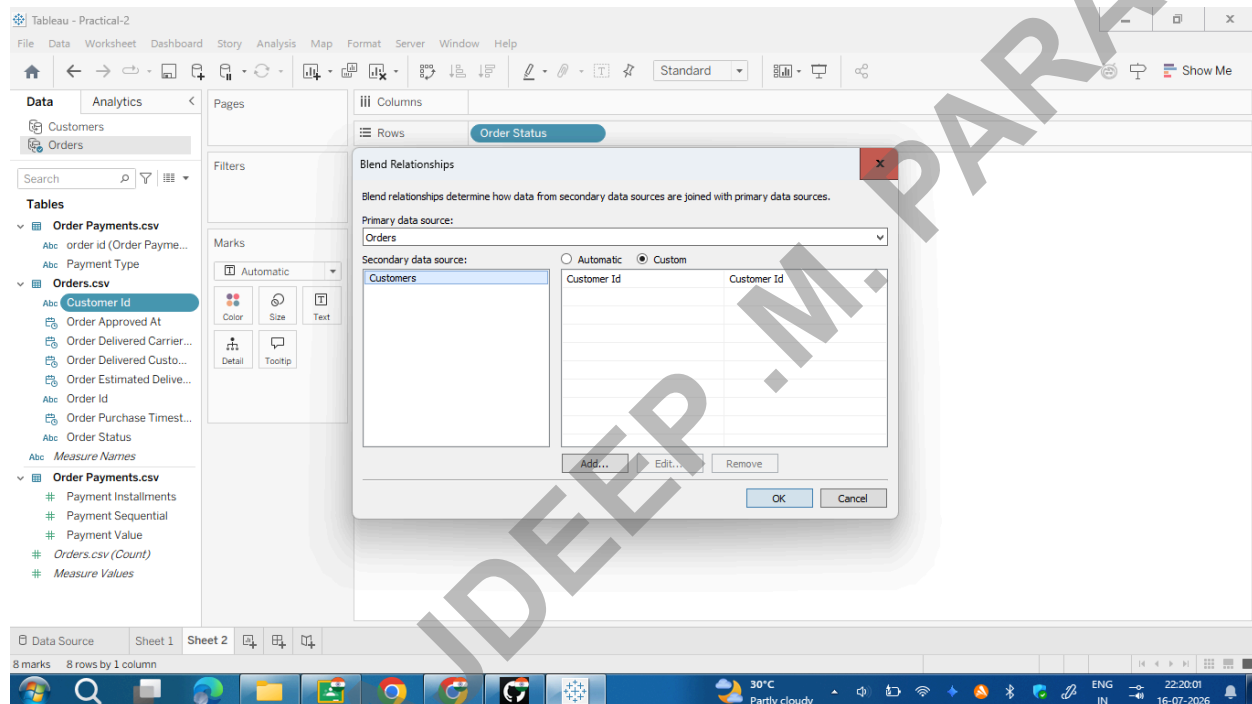
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Now Go Back to any to the Sheet 2 and right click on the Data Pane's empty space and click on "Show Hidden Fields"

After Clicking it the Customer Id comes back into the data pane and make sure you click the "eye button" next to it marked in bellow provided snapshot.

Now go back to Edit Blend Relationship option and after clicking it you may see both Customer Id in Left and Right, After that click on Custom it changes from automatic to custom and click Ok

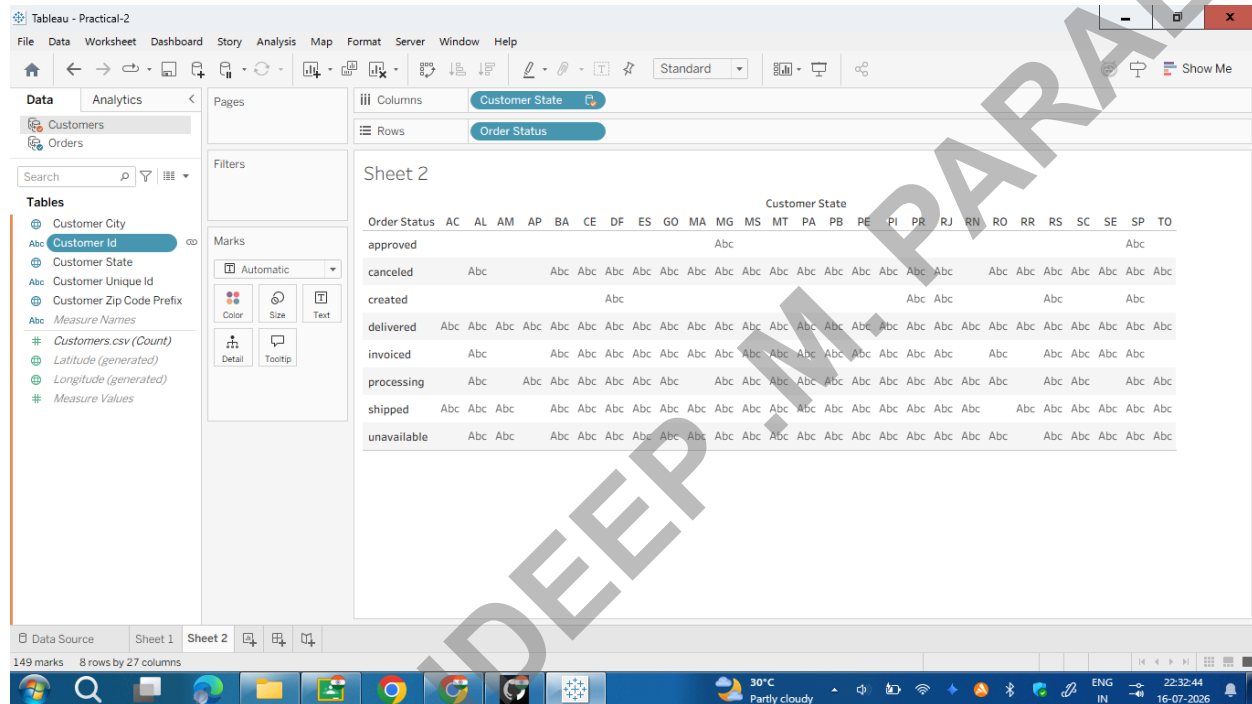


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6. Now, go back to the top-left section of your Data Pane and click directly on the separate Customers dataset tracking label. An orange chain link icon will now appear locked together immediately next to the customer_id field.

At the start you may not see the orange tick on the Customer database for that click on the unlinked chain next to the customer id on the data pane after clicking the orange tick appear as shown in the snapshot

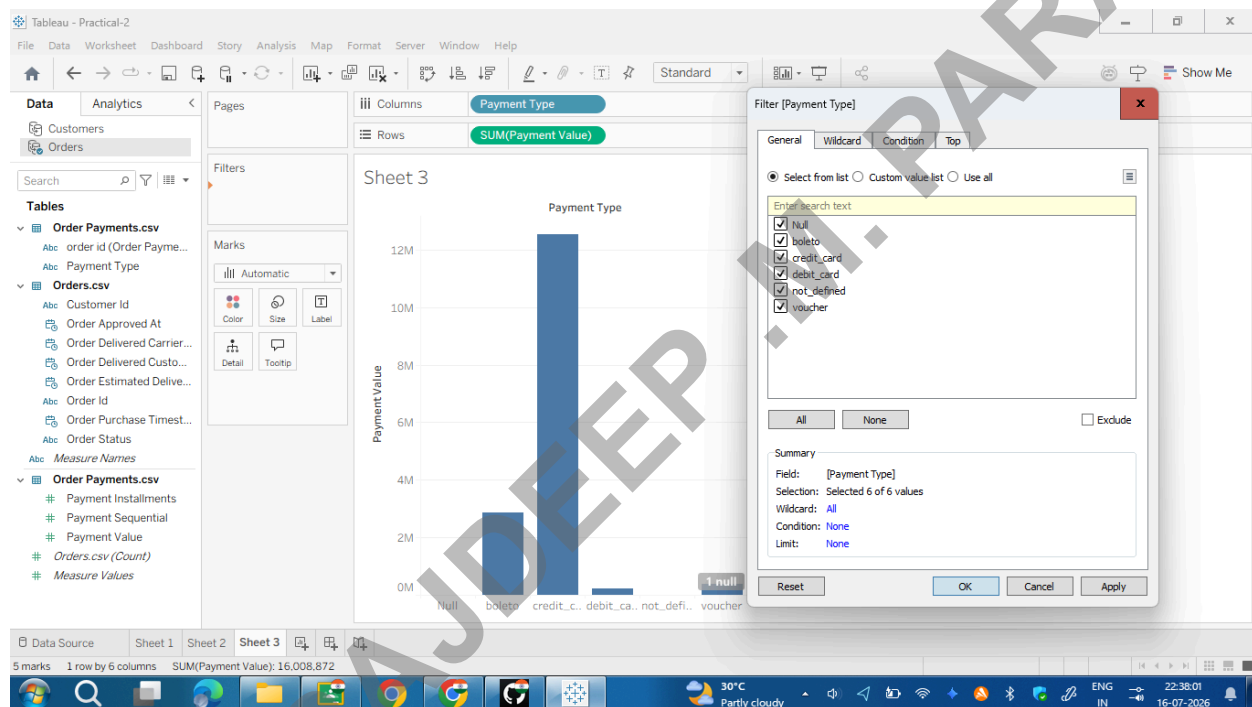


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Step F: Implementing Multi-Tiered Filters

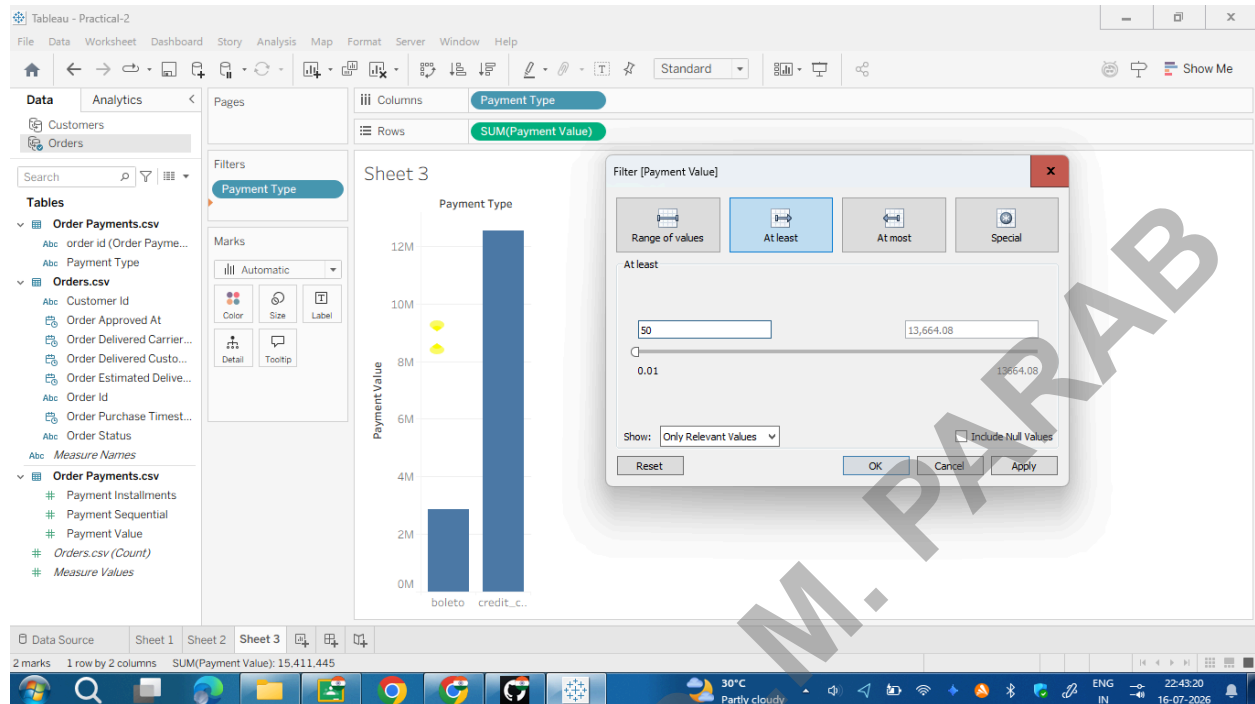
1. Open a clean worksheet. Drag the quantitative measure field payment_value onto the Rows shelf and place the categorical dimension payment_type onto the Columns shelf.
2. Dimension Filter: From your Data Pane list, drag the categorical string attribute payment_type and drop it onto the empty Filters shelf positioned immediately above the Marks card layout. Uncheck the box for not_defined, check the boxes for credit_card and boleto, and click OK.



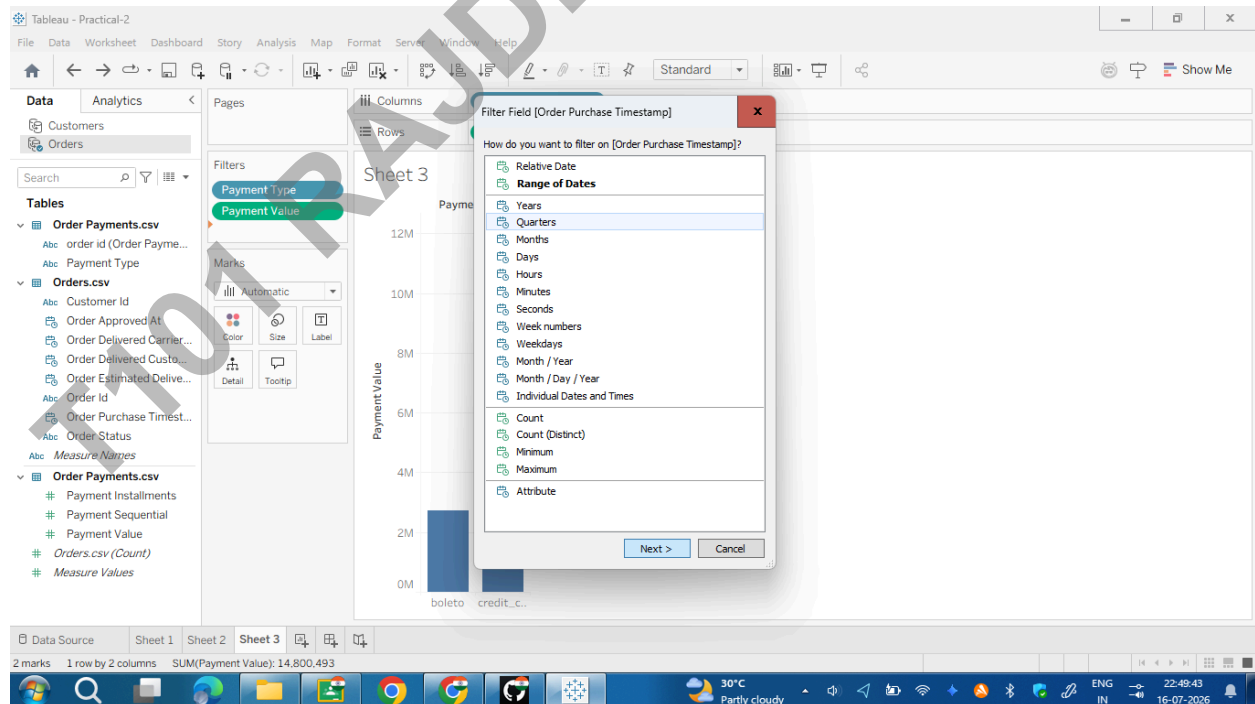
3. Measure Filter: Drag the quantitative numeric variable payment_value out from the measures list and drop it directly onto the Filters shelf. Select All Values in the prompt window, click Next, select the At Least option condition, type in 50 to screen out microtransactions(it will be at 0.1 at by default Change it by clicking on it and typing 50 on it), and click OK.

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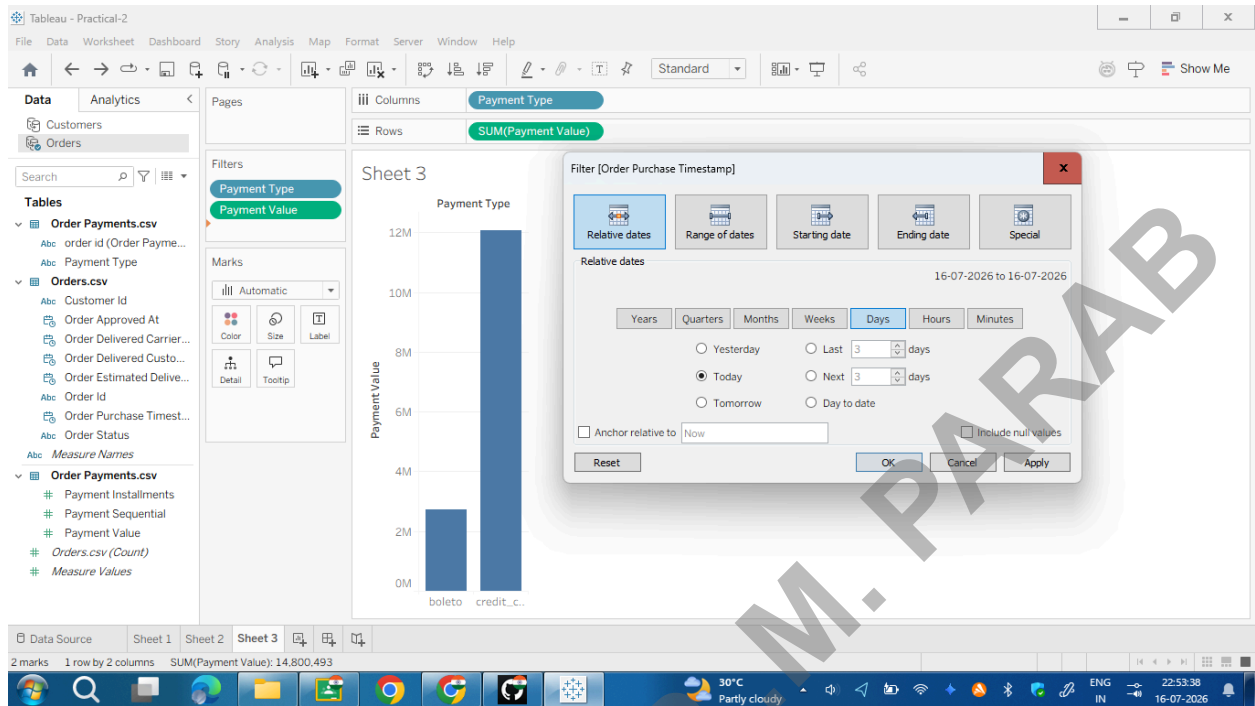


4. Date Filter: Drag the calendar configuration time variable `order_purchase_timestamp` and drop it onto the Filters shelf. Select Years or Relative Date from the prompt configuration layout page to isolate the visualization into specific historical reporting boundaries .



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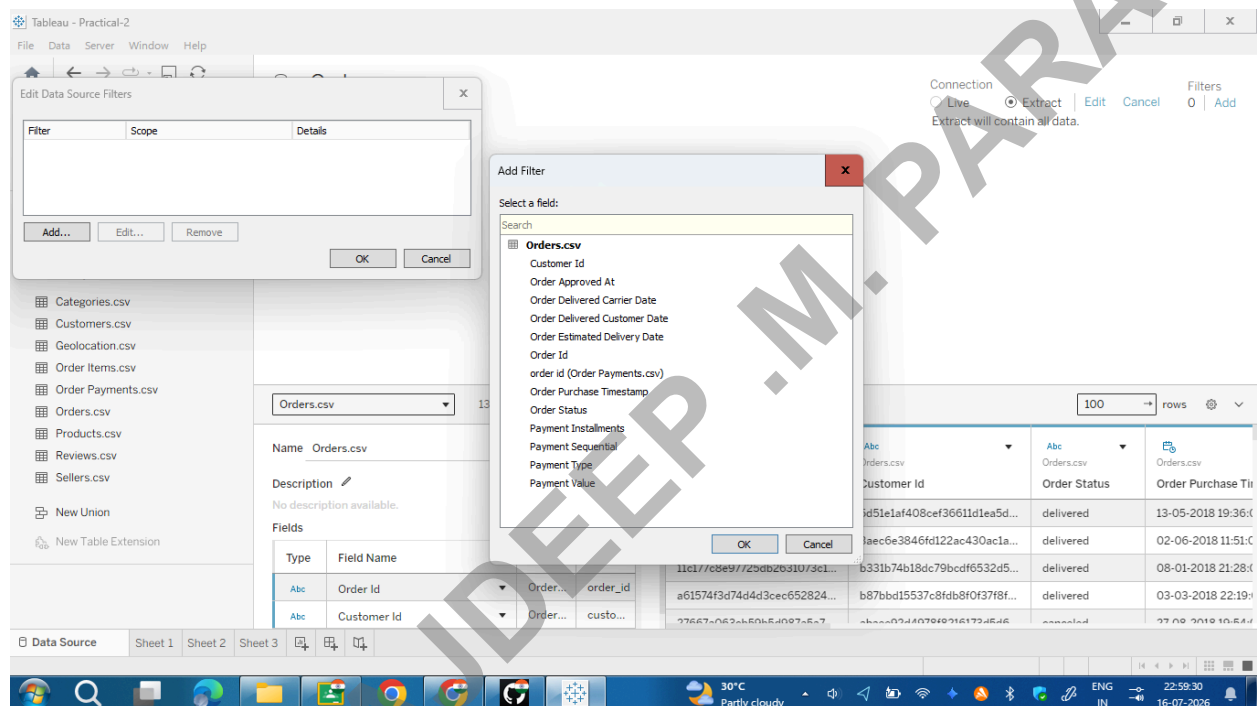


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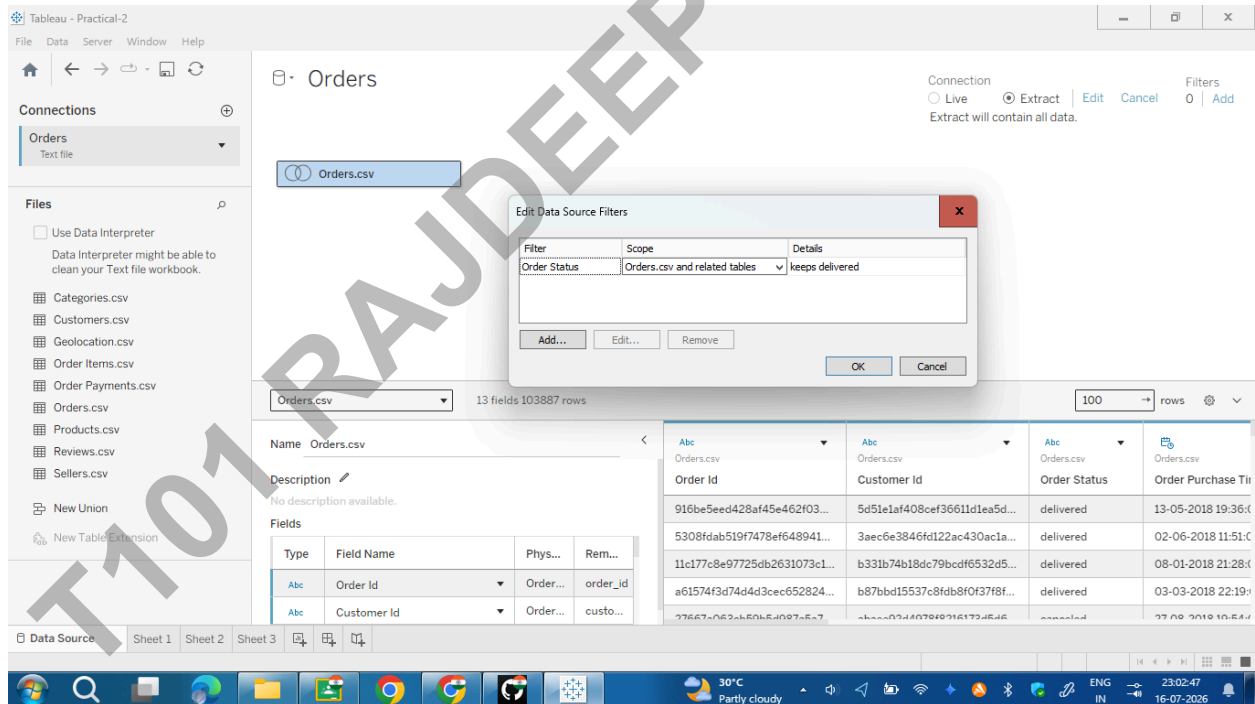
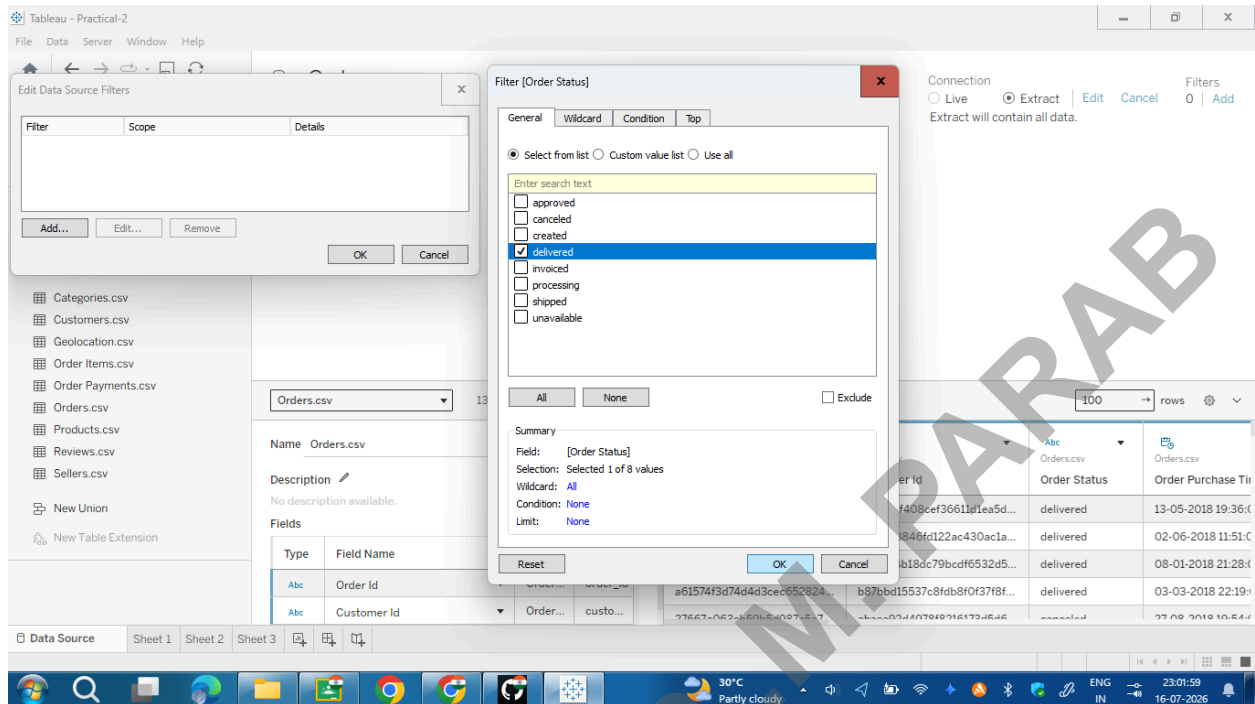
Step G: Optimizing Dashboard Performance

1. Extract Filters: Navigate back to the main Data Source configuration workspace screen. Look immediately below the selected Extract radio dial selection button in the upper-right area and click the small Edit link. Click Add, select the categorical field Status, choose to check delivered exclusively, and click OK. Tableau will purge all canceled or unfulfilled order rows from the local extract file entirely



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2. Context Filters: Return to an active worksheet layout canvas view. Move your cursor to your Filters shelf, right-click any active filter pill sitting in the shelf area, and select Add to Context from the context dropdown. The selected pill background formatting will immediately change to a dark gray color.

